

ForgeCloud Backend & API Design

FastAPI control plane, services, APIs, authentication and lifecycle

1. Backend Responsibility

FastAPI is the ForgeCloud control plane. It provides REST APIs, authentication, application lifecycle logic, deployment orchestration interfaces, infrastructure operations, monitoring endpoints and audit logging.

2. Backend Architecture

```
FastAPI
|
+-- API Routes
|   +-- auth
|   +-- users
|   +-- applications
|   +-- deployments
|   +-- infrastructure
|   +-- monitoring
|   +-- logs
|   +-- audit
|
+-- Services
|   +-- Auth Service
|   +-- Application Service
|   +-- Deployment Service
|   +-- Infrastructure Service
|   +-- Monitoring Service
|
+-- Models / Schemas
|
+-- Database Layer
|
+-- External Integrations
|   +-- Terraform
|   +-- GitHub
|   +-- Kubernetes
|   +-- AWS
```

3. API Surface

Method	Endpoint	Purpose
POST	/api/auth/login	Authenticate user
GET	/api/users/me	Current user
GET	/api/applications	List applications
POST	/api/applications	Create application
GET	/api/applications/{id}	Application details
PUT	/api/applications/{id}	Update application
DELETE	/api/applications/{id}	Delete application
POST	/api/applications/{id}/deploy	Trigger deployment
POST	/api/applications/{id}/rollback	Rollback
GET	/api/applications/{id}/deployments	Deployment history

Method	Endpoint	Purpose
GET	/api/infrastructure	Infrastructure status
POST	/api/infrastructure/provision	Provision infrastructure
GET	/api/metrics	Monitoring data
GET	/api/logs	Application/platform logs
GET	/api/health	Health status
GET	/api/audit	Audit records

4. Authentication and RBAC

ADMIN

full platform access

DEVELOPER

create/manage applications
deploy/rollback
view operational data

VIEWER

read-only access

5. Application Lifecycle

CREATE

-> CONFIGURE
-> BUILDING
-> DEPLOYING
-> RUNNING

Failure:

DEPLOYING
-> FAILED
-> ROLLED_BACK

6. Backend Engineering Requirements

- Validate all incoming request data.
- Never store plaintext passwords.
- Use authorization checks on every protected operation.
- Keep infrastructure credentials outside source code.
- Return consistent HTTP status codes and error payloads.
- Log operational events without leaking secrets.
- Separate API routes, business logic and database access.
- Write unit and API integration tests.